A-A-50139
December 31, 1987

COMMERCIAL ITEM DESCRIPTION

ROPE, KERNMANTLE

The General Services Administration has authorized the use of this commercial item description.

1. Classification. This commercial item description covers nylon kernmantle ropes in the following types.

Type I - Kernmantle rope shall be 7 mm in diameter and 150 feet (40 m) long in uninterrupted length. The minimum breaking strength shall be 900 kg.

Type II - Kernmantle rope shall be 8 mm in diameter and 150 feet (40 m) long in uninterrupted length. The minimum breaking strength shall be 1250 kg.

Type III - Kernmantle rope shall be 9 mm in diameter and 150 feet (40 m) long in uninterrupted length. The minimum breaking strength shall be 1575 kg. The kernmantle rope shall be approved by the U.I.A.A. (Union Internationale des Associations d'Alpinisme) to hold a minimum of 7 U.I.A.A. (see 4.3) falls and impacts a minimum force of 750 deka Newton (daN).

Type IV - Kernmantle rope shall be 11 mm in diameter and 165 feet (45 m) long in uninterrupted length. The minimum breaking strength shall be 2400 kg. The kernmantle rope shall be approved by the U.I.A.A. (Union Internationale des Associations d'Alpinisme) to hold a minimum of 9 U.I.A.A. (see 4.3) falls and impacts a maximum force of 1200 daN on a test weight. The kernmantle rope shall be water repellent treated to resist water penetration into the rope. The maximum average dynamic absorption shall be 26.5 percent.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The kernmantle rope shall be nylon in construction consisting of a braided outer nylon sheath (mantle) and an inner nylon core (kern). All ends of kernmantle ropes shall be cut with a hot knife and taped to prevent fraying. Mantle color shall be black.

FSC 4020

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3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify that the product offered meets the salient characteristics of the description, and conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same as the product offered for sale in the commercial marketplace. When specific quality assurance provisions are specified for any commercial characteristic, the contractor shall furnish data resulting from inspection conducted in accordance with the specific quality assurance provisions. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Testing. Types I through IV kernmantle ropes shall be tested for breaking strength in accordance with test method 6016 of Federal Standard No. 191. Minimum breaking strengths are listed above in the salient characteristics.

3.3 Type IV rope impact test. The type IV rope, in addition to the above test, shall be tested as follows to determine how many U.I.A.A. falls it will sustain without breaking. This test shall also determine the impact force on the weight. An 80-kg weight is dropped from a height of 5 meters. It is held by a 2.8-meter length of rope passed over a 10-mm carabiner edge. Two measurements are taken: the number of falls held (five minimum) and the impact force on the weight (1200 kg maximum). Failure of the rope to withstand five falls or failure to meet the 1200-kg minimum impact force requirement shall constitute failure of the test. Any industrial testing machine may be used to determine compliance with this requirement.

3.4 Type IV rope dynamic absorption test. Type IV rope shall be tested for dynamic water absorption in accordance with test method 4500 of Federal Standard No. 191, "Textile Test Methods," except for the following changes to the method:

1. A hot knife (120 Volt Unger #7760 or equivalent, with a 5/16 inch flat tip heating element) shall be used to cut the specimens and seal the rope to prevent unraveling of the ends.

2. In lieu of using a wringer, at the end of the required running time, one specimen shall be grasped at one end and the opposite end tapped downward against a solid object four times. The specimen shall be turned 180°, grasped at the opposite end, and similarly tapped four times at the point previously held. The tapping procedure shall be repeated one additional cycle for each end.

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The specimen shall immediately be placed (all pieces laid side to side) smoothly between two sheets of blotting paper. The specimen and blotting papers shall be firmly hand-pressed for not more than 5 seconds.

The specimen shall be left between the two blotters for a minimum of 3 minutes and a maximum of 4 minutes.

4. PACKAGING

4.1 Packaging. Each length of rope shall be coiled, flattened and tied into hanks not exceeding 24 inches in length. Each coiled length of rope shall then be unit packed in a snug-fitting 200-pound test corrugated fiberboard box or a sealed polyethylene bag with a wall thickness that exceeds 0.003 inches.

4.2 Packing. Four hanks of rope of one type, packaged as specified, shall be packed in a close-fitting RSC style corrugated fiberboard shipping container conforming to type CF, class domestic, grade 275 of PPP-B-636. The shipping container shall be completely filled, without voids, and of a size that can be palletized in accordance with MIL-STD-147. Each shipping container shall be closed in accordance with the appendix of PPP-B-636.

4.3 Palletization. When specified (see 5.2), rope packed as specified shall be palletized on a 4-way entry pallet in accordance with load type 1a of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L, or film bonding O or P, in accordance with MIL-STD-147.

4.4 Marking. In addition to any special markings required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

5. NOTES

5.1 Intended use. The kernmantle rope is intended to provide security for men or equipment in operations involving steep ascents. It is also intended to be used for hauling equipment.

5.2 Ordering data.

a. Title, number, and date of this document.
b. Type required (see 1.1).
c. When palletization is required (see 4.3).

5.3 Source of Information for nongovernmental document.

Union Internationale des Associations d'Alpinisme (UIAA)
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